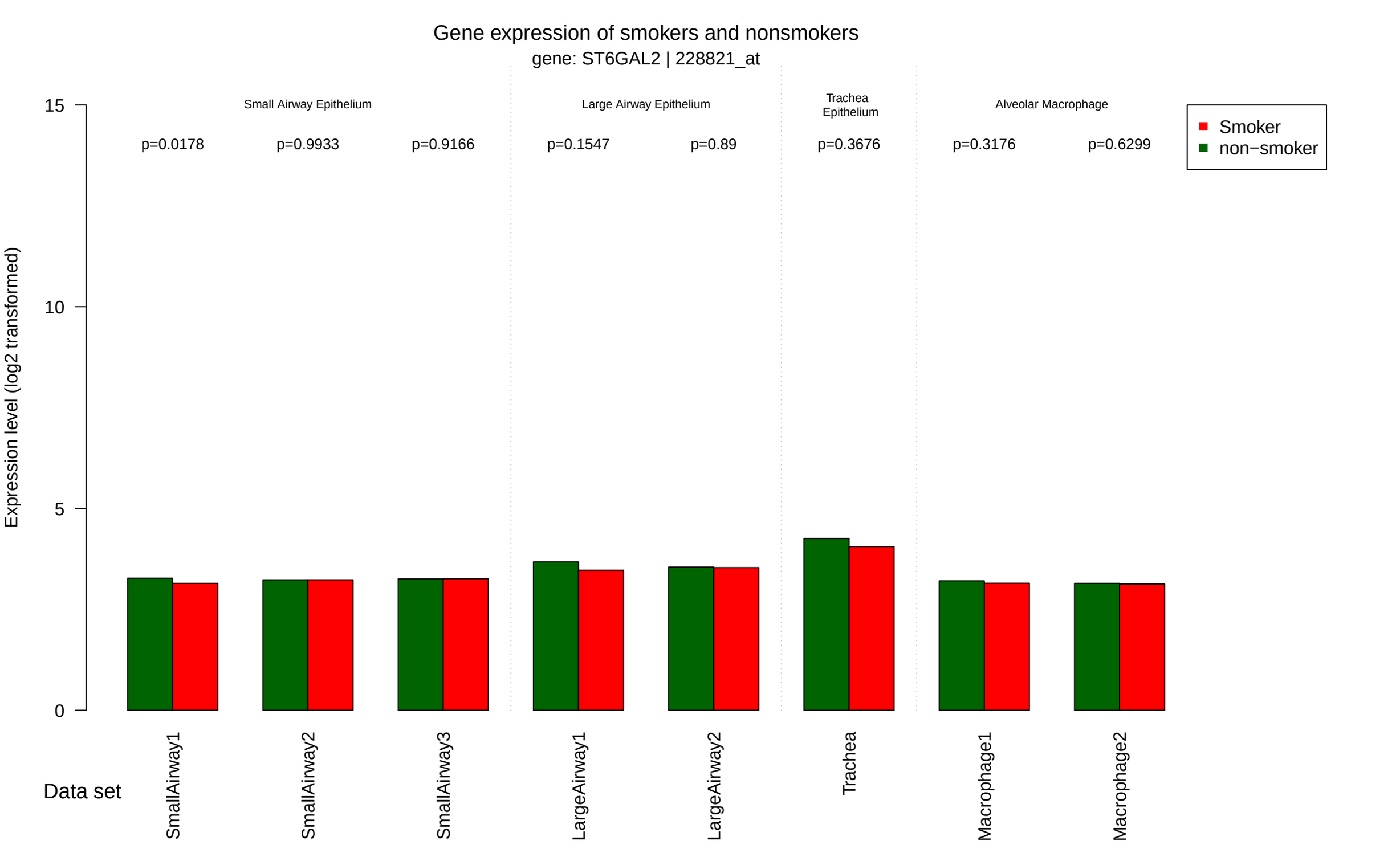
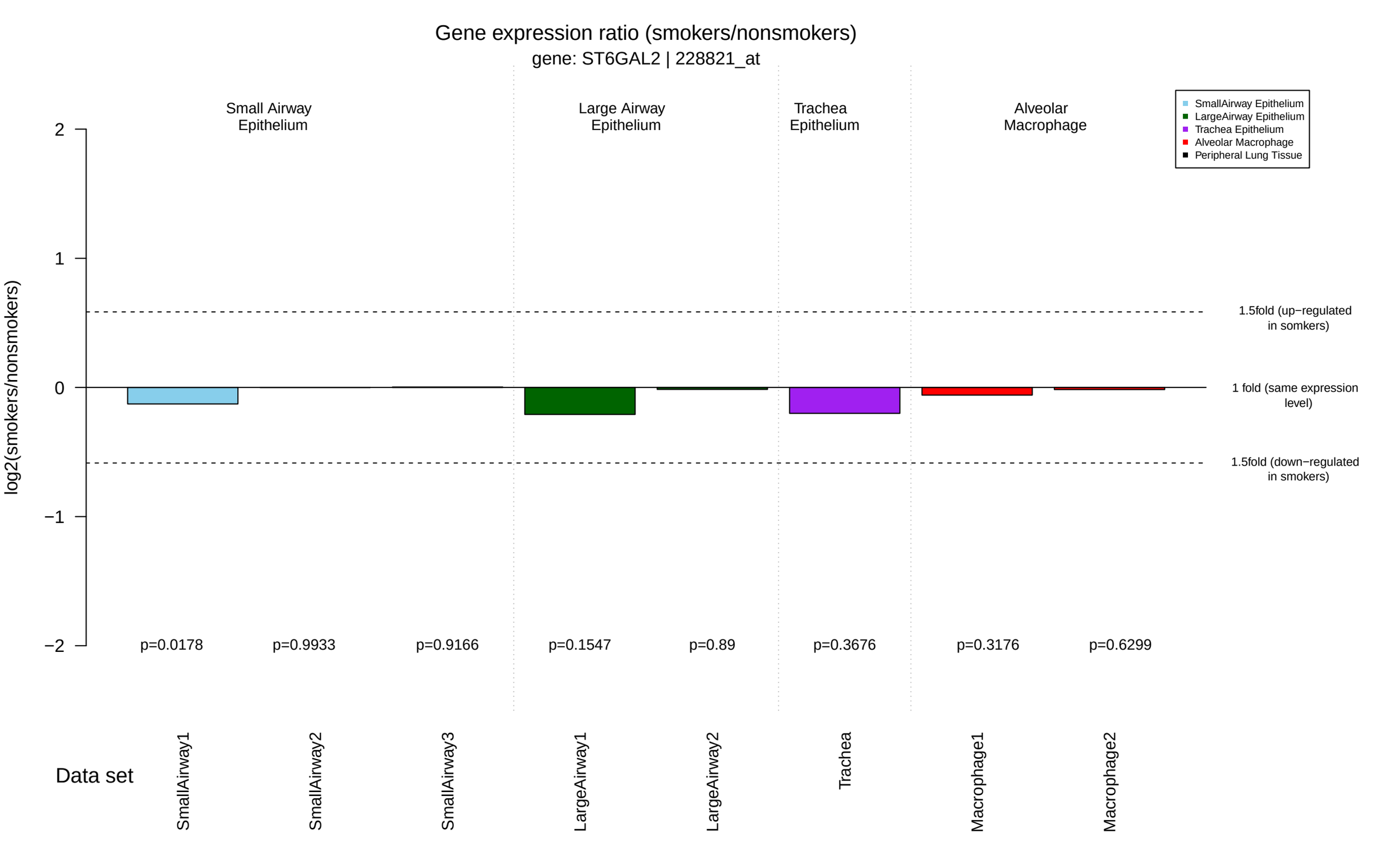
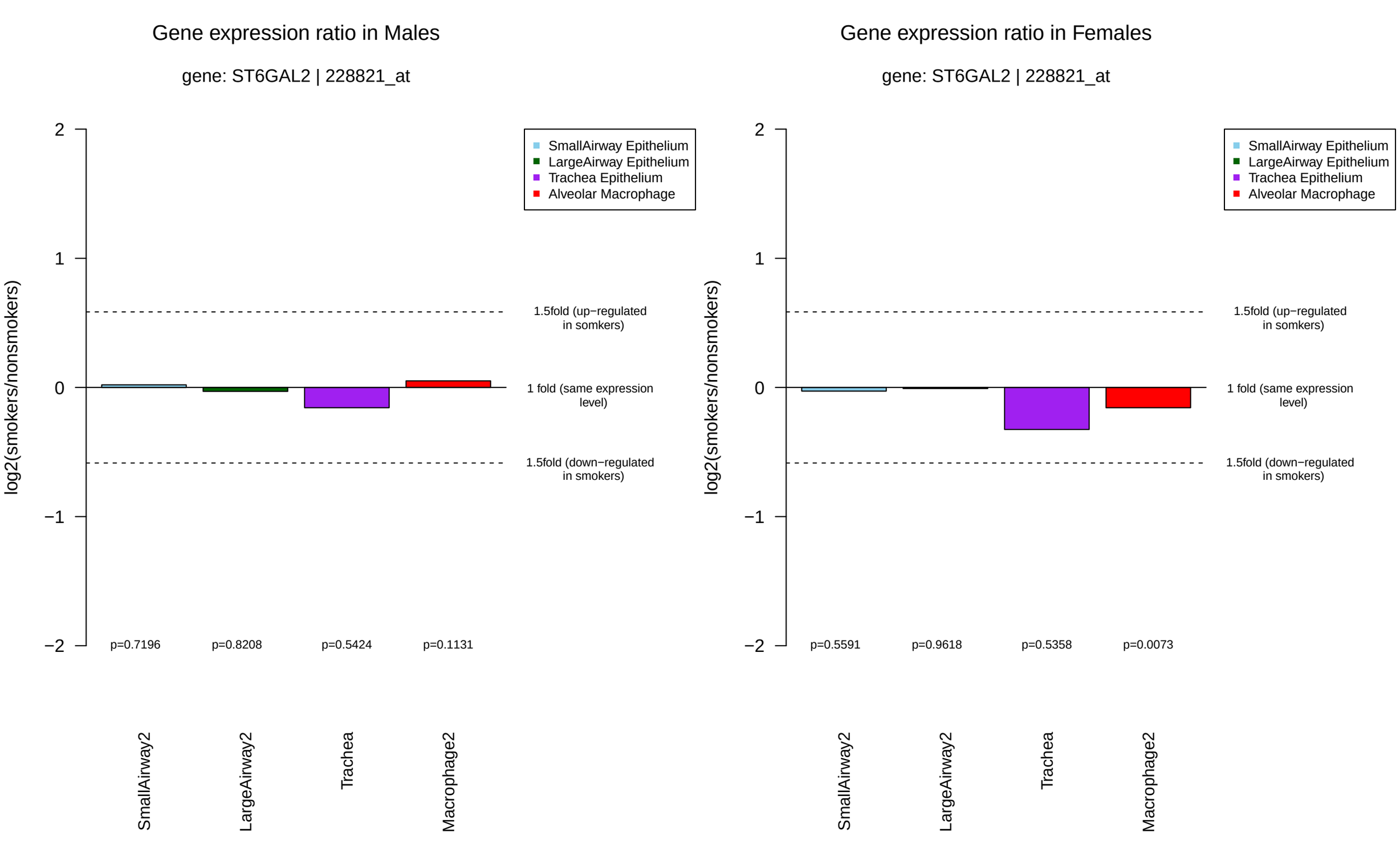
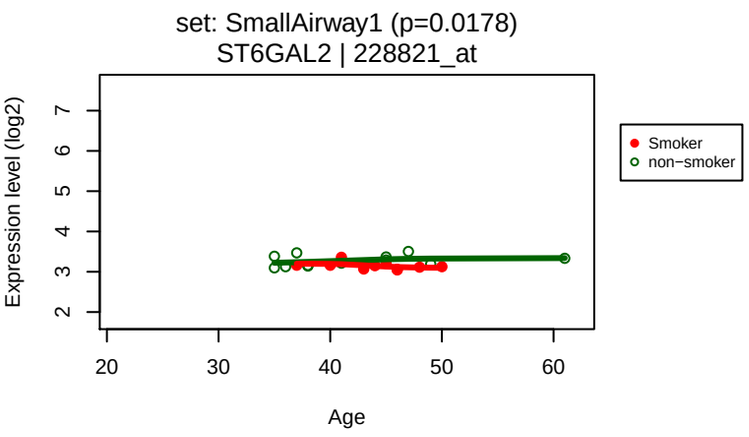
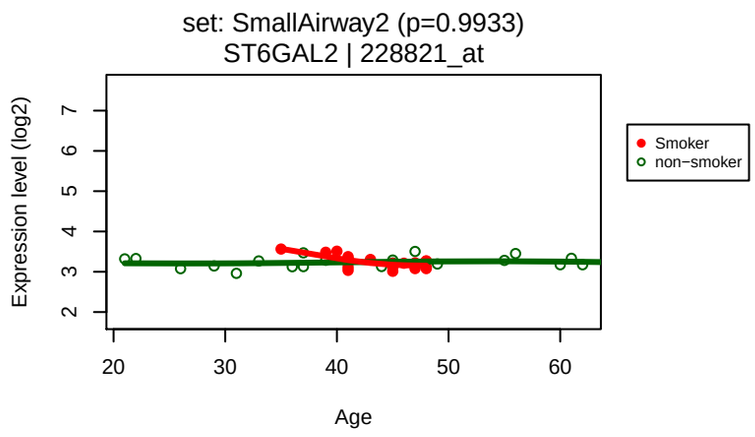
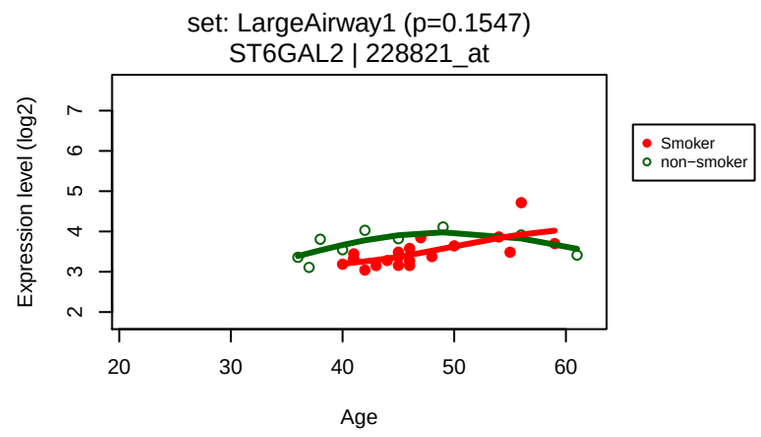

Gene expression of smokers and nonsmokers
gene: ST6GAL2 | 228821_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0178
p=0.9933
p=0.9166
p=0.1547
p=0.89
p=0.3676
p=0.3176
p=0.6299
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

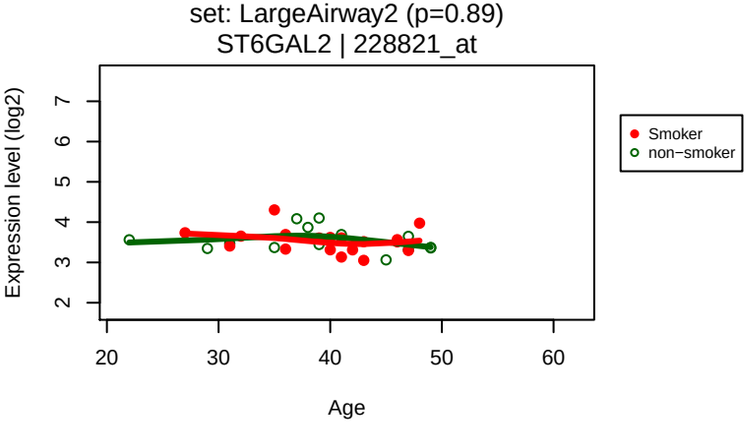

Gene expression ratio (smokers/nonsmokers)
gene: ST6GAL2 | 228821_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0178
p=0.9933
p=0.9166
p=0.1547
p=0.89
p=0.3676
p=0.3176
p=0.6299
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

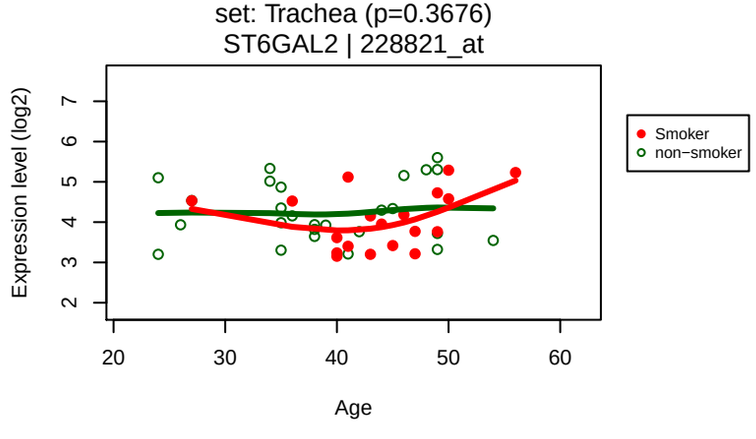

Gene expression ratio in Males
gene: ST6GAL2 | 228821_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7196
p=0.8208
p=0.5424
p=0.1131
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST6GAL2 | 228821_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5591
p=0.9618
p=0.5358
p=0.0073
SmallAirway2
LargeAirway2
Trachea
Macrophage2

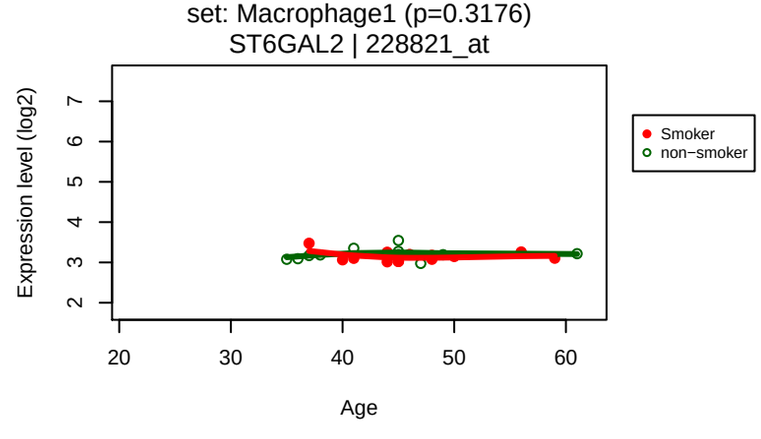

set: SmallAirway1 (p=0.0178)
ST6GAL2 | 228821_at
Expression level (log2)
Age
Smoker
non−smoker

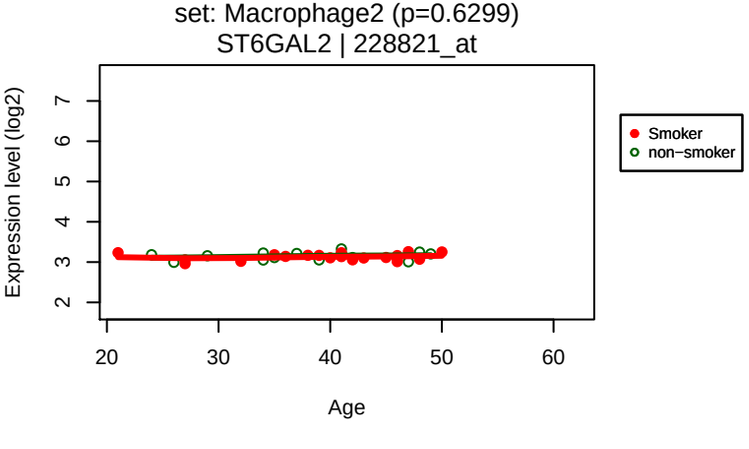

set: SmallAirway2 (p=0.9933)
ST6GAL2 | 228821_at
Expression level (log2)
Age
Smoker
non−smoker

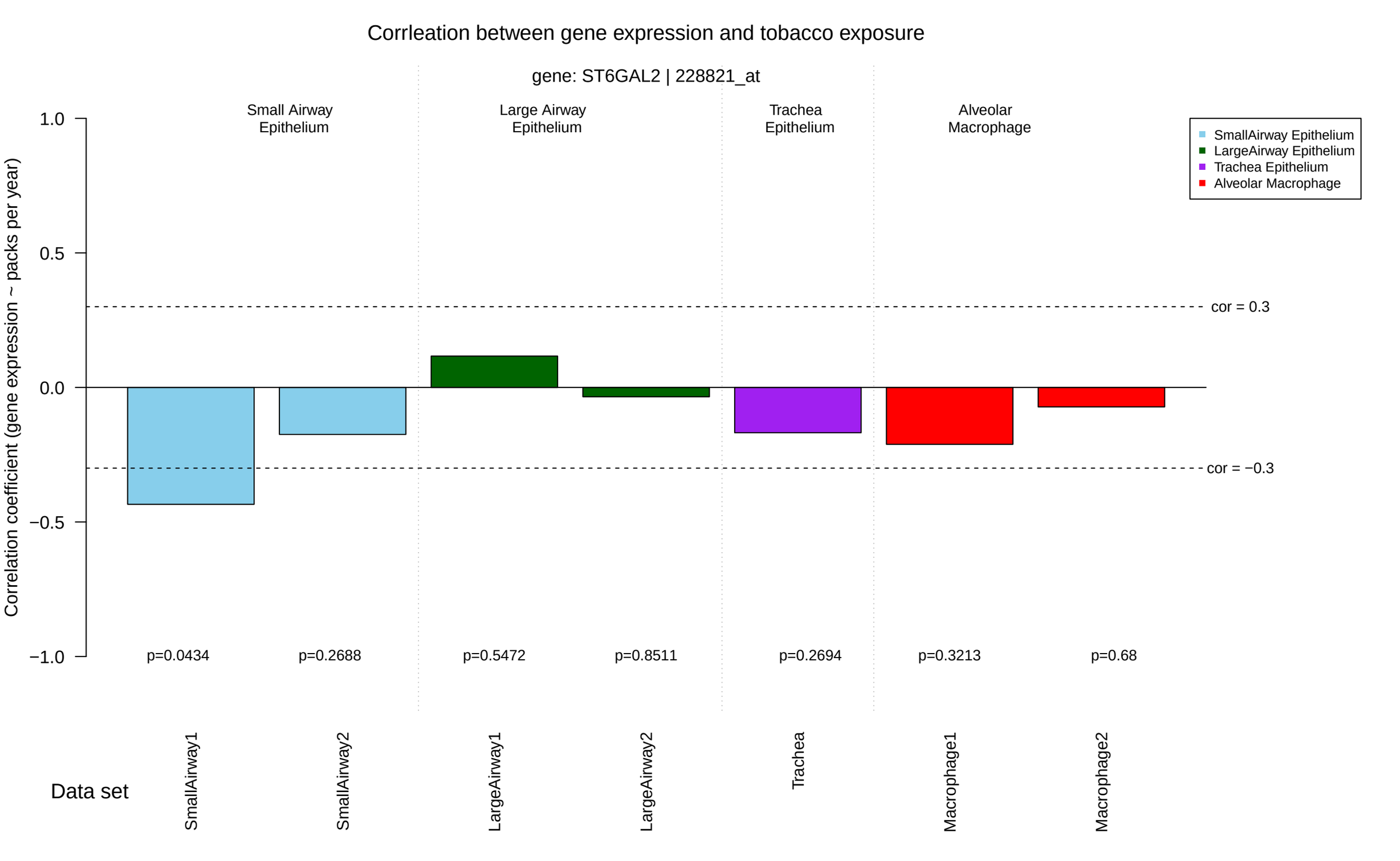

set: LargeAirway1 (p=0.1547)
ST6GAL2 | 228821_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.89)
ST6GAL2 | 228821_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3676)
ST6GAL2 | 228821_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3176)
ST6GAL2 | 228821_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6299)
ST6GAL2 | 228821_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST6GAL2 | 228821_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0434
p=0.2688
p=0.5472
p=0.8511
p=0.2694
p=0.3213
p=0.68
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2